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| **PI Name:** | Su, Ching-Hua Ph.D. | | |
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| **No. of Bachelor's Candidates:** | 0 | **Monitoring Center:** | NASA MSFC |
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| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | Lehoczky, S. ( NASA MSFC )<br>Scripa, R. ( University of Alabama at Birmingham )<br>Ban, Heng ( University of Alabama at Birmingham )<br>Benmore, Chris ( Argonne National Laboratory ) | | |
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| **Task Description:** | We propose to to perform ground-based fundamental study of the thermophysics and crystal growth of Cd1-xZn xTe (0.1<x<0.2) materials suitable for high-energy radiation sensors to obtain a detailed understanding of the origins and mechanisms of defect formation and to conduct ground-based experimental and theoretical research on the thermophysical and thermodynamic properties and structural fluctuations of III-VI and I-III-VI solids and melts to study the fundamental heterophase fluctuations phenomena in the melts. The specific tasks are detailed below: 1) Thermophysical properties measurements of CdZnTe solids and melts. 2) Crystal growths of CdZnTe using the techniques of gradient freeze and vapor transport. 3) Characterization of structural quality and electrical properties of grown CdZnTe. 4) CdZnTe device fabrication and testing. 5) Conducting neutron scattering analysis and quantitative measurements of the relevant thermophysical properties of the III-VI and I-III-VI melts such as viscosity, electrical conductivity, thermal diffusivity and density as well as the relaxation characteristics of these properties to advance the understanding of the structural properties and the relaxation phenomena in the melts 6) Performing partial pressure measurements of Se2, Te2, and other vapor species and obtaining the enthalpy of mixing for Se and Te, liquidus temperature, and homogeneity range of the III-VI and I-III-VI solid phase. 7) Theoretically, an associated solution model for the liquid phase will be generalized to treat a liquid consisting of any type and number of species suggested by previous results and the neutron scattering analysis. Phase diagram calculation and defect chemistry analysis will be performed to interpret the experimental results. The systems to be studied include In-Se, Ga-Se, In-Te, Ga-Te, Ag-Ga-Se, and Ag-Ga-Te. The project will be a joint study including participants from NASA/Marshall Space Flight Center, Fisk University/NASA Center for Photonic Materials and Devices (CPMD), University of Alabama at Birmingham (UAB), British Aerospace Engineering (BAE) and Argonne national Laboratory. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | The basic properties and the growth process of CdZnTe result in the uncontrolled formation and evolution of structural defects and non-uniformities during crystallization and the subsequent cool-down that inhibits efficient charge collection when an external field is applied. This inevitably degrades the energy resolution and overall response uniformity, and forces users to compromise with rather small (~ few cm2 x mm) devices. Also, the expense of raw materials (currently about $1/g), the typically low yield from the growth process (about 10 to 20%), i.e., poor quality ingots, and the need to attach complicated electrode structures to compensate for bulk and surface effects on charge collection and noise, have kept the unit cost of CdZnTe detectors prohibitive for wide-field instruments like EXIST and ACT (the price for SWIFT was about $1.75/mm3) and other applications. Thermal and electrical conductivity, viscosity and density are basic thermophysical properties of semiconducting melts. The experimental values of the properties as functions of temperature and composition are required for virtually all melt systems for microgravity materials processing as well as in most areas of fluid physics. The database of the equilibrium values of the properties is incomplete and data on their non-equilibrium behavior of the properties are scarce or non-existent. Accurate data for both of these cases as functions of temperature and composition would significantly increase the quality of solidification and crystal growth research and would make numerical modeling more meaningful. For example, accurate data on thermal conductivity for the ternary solid and liquid systems as functions of both temperature and composition are prerequisite for numerical simulation of the evolution of the solid-liquid interface during the solidification. Beyond utility in modeling of crystal growth, accurate benchmark data for the thermodynamic and thermophysical properties are extreme helpful for a theoretical elucidation of the phenomena of structural transformations and heterophase fluctuations for a large class of complex semiconducting liquids having group VI elements as one of their main components. |
| | I. Cadmium Zinc Telluride (CdZnTe) High-Energy Radiation Sensors<br>One of the most critical requirements for CdZnTe in the application of room temperature nuclear radiation detector is high electrical resistivity. This is paramount in order to reduce the bulk leakage current to an acceptable level. The high resistivity can only be achieved with materials of low carrier concentrations, which are controlled by the concentrations of intrinsic defects, i.e., native point defects such as Cd vacancy, and extrinsic point defects, including intentional and unintentional dopants. The information on the solidus points and their equilibrium vapor pressures is therefore an important parameter in the control of the concentration of native point defects. As a part of our initial investigation, we have performed the partial pressure measurements by optical absorption technique and established the corresponding xTe-T-PTe2 data on the Te-rich solidus for Cd0.80Zn0.20Te. This was the first comprehensive thermodynamic characterization performed for the CZT crystal system. It was performed in the materials science and crystal growth laboratories of NASA/MSFC and expected to be of significant value to CZT crystal producers.<br><br>If one prepares starting materials of stoichiometric composition, i.e., xTe = 0.5000, it will lose more Cd to the vapor phase than Te and the composition of the condensed phase will shift toward Te-rich. There are two major consequences of the shift: Cd vacancy and Te-rich second phase. Firstly, the native defect associated with a Te-rich crystal is believed to be mainly Cd-vacancy. The Cd-vacancy is doubly-ionized acceptor and can affect the electrical properties of the CdZnTe crystals tremendously. Secondly, with the usually slow cooling rate after the growth, the solidified crystal will consists of a mixture of CdZnTe solid and a Te-rich second phase because of the retrograded nature of the solidus curve. By applying the thermodynamic information of CdZnTe system, one can reproducibly control the stoichiometry of the melt and, consequently, the stoichiometric composition of the solid that grows from it.<br><br>The starting materials of CdZnTe were homogenized from pure elements of Cd, Zn and Te in a rocking furnace. However, it was found from chemical analyses that the first two grown crystals were contaminated by Cu, which presumably came from either the attached thermocouples or from the furnace heating elements. Therefore, for the growth ampoules after the third run the ampoules were homogenized inside the rocking furnace under vacuum condition by inserting a larger fused silica closed tube outside the sample ampoule and maintaining a vacuum condition inside the closed tube during homogenization.<br><br>During the investigation so far, about forty Cd0.80Zn0.20Te crystals have been grown using directional solidification by either vertical gradient freeze or Bridgman techniques. To control the stoichiometry of the condensed phase, a Cd reservoir on the top of the melt was used to control the Cd over-pressure. The range of the Cd reservoir temperature was from 750 to 935oC, with the range of 785 to 820oC for most of the runs. Both seeded and unseeded growth ampoules were processed. The ampoule sizes were either 20mm ID (25mmOD) or 40mm ID (45mm OD). The translation rates were 0.75 to 2mm/h. After the translation was completed the sample and the Cd reservoir were cooled down to room temperature over a period of 96 to 144 hours. For about fifteen of the growth runs mentioned above a small amount of In dopant was added to the starting materials during the homogenization process in the rocking furnace. |

**Task Progress:**

Using a Cd reservoir for the precise control of stoichiometry as well as proper doping of the starting material, high resistivity (above 1010 Wcm) crystals of CdZnTe have been reproducibly grown. Also, by experimenting with various growth parameters and sample ampoule designs, single crystal has been consistently produced on the first 70% of a 20mm diameter ingot. For the larger diameter (40mm) samples, the crystals usually consisted of two major grains for the first 70% of the ingots. Twinning was observed to be confined in the shoulder region of the first grown section. Occasionally, completely twin-free ingots were grown and the causes are being studied.

The samples produced were characterized by the following methods:

i. Microprobe analysis:

The compositional uniformity of the grown crystals was scanned by microprobe and the composition of ZnxTe established. The pseudobinary nature of the CdTe-ZnTe system resulted an axial compositional profile starting with high x, about 0.24, which decreased gradually to 0.18 at the top. A radial distribution of a wafer sliced perpendicularly to the growth axis shows that compositional uniformity was about +0.007.

ii. Glow discharge mass spectrometer (GDMS):

The impurity contents in the grown crystals were examined by GDMS. As mentioned above, the results show that the first two grown ingots were contaminated by Cu (3300ppb). After the modifications of procedure, i.e. homogenized under vacuum and no thermocouples attached to the ampoules, the GDMS results show that the Cu content has been reduced to 130ppb. The data also show that the amount of major impurities such as C, O and Mg were also reduced.

iii. IR transmission images and microscope:

The IR transmission images were taken on large CdZnTe pieces or sliced wafers by an IR camera and IR microscope to study the structural defects, such as cracks, stress concentrators, twins as well as Te (or Cd) precipitates.

iv. Hall measurements:

Considering the high resistivity of the CZT crystals, we have tried various electrical contacts, including indium solder, gold epoxy paste, silver paste, Au-Cl hydrate and Au thin film. Others contact types will be tried in the future work. It was found that for the most reliable ohmic contacts, the samples were prepared by sputtering a 3mm Au film with indium solder on top for the lead wires.

Hall measurements were performed from room temperature to 78K with a field strength of 10KGauss. Two quantities, resistivity and Hall coefficient, were measured. Their values are determined by four parameters, namely, electron concentration, hole concentration, electron mobility, and hole mobility.

A computer code has been updated for the computation of the charge-carrier concentrations, Hall coefficients, Fermi energies, and electron mobility for Cd1-xZnxTe as functions of x, temperature and ionized-defect and neutral defect concentrations. The electron mobility was calculated in terms of a microscopic theory of electrical conduction derived from the solution of the Boltzman equation for the perturbed steady-state electron distribution function as has been described previously. The calculated value at 295K was 1100 cm2/V-s and the hole mobility, mp, have been reported for CdTe as 1/10 of that of electron. Using the calculated value for electron mobility and 1/10 of that as hole mobility, the measured resistivity and Hall coefficient, the individual values of electron and hole concentration can be determined. The temperature dependence of the electrical properties have been measured between 7 to 300K for a series of wafers cur from various ingots to estimated carrier types and concentrations. The results show that, in these high resistivity samples, the electron and hole concentrations are in the range of 105 to 107cm-3, or about 10-8 to 10-6 ppb, atomic. Without a very precise control of the donor and acceptor concentration and compensation between them, it is almost impossible to reproducibly grow high resistivity crystals. Results also show that the sample CZT-29a is n-type (n > p) and the other two are p-type (p > n).

Status of Present CZT Crystal Growth Research at MSFC

The proposed goal of developing technology to enhance the yield of large CdZnTe crystals for high-energy radiation detectors has been partially accomplished. Our approach in the previous research program was to consistently produce large single crystals of high crystalline quality by conducting systematic crystal growth experiments, followed by a study of the electrical properties so as to consistently grow CdZnTe crystals of high electrical resistivity, in undoped crystals.

We have accomplished the following: (1) established thermodynamic properties of the CdZnTe system (2) developed a procedure for homogenizing the pure elements by a localized eutectic reaction using a traveling zone process (3) eliminated the Cu and other elemental contamination by homogenizing the pre-growth CdZnTe mixture under vacuum condition (4) consistently grew high resistivity CdZnTe crystals by controlling the Cd over-pressure and dopant concentration, and (5) developed a growth process, which is capable of producing large grain, high crystalline quality CdZnTe.

Publications:

1. Ching-Hua Su, “Composition – Temperature – Partial Pressures Data for Cd0.8Zn0.2Te by Optical Absorption Measurements” J. Crystal Growth 281 577-586 (2005).

2. Ching-Hua Su, S. L. Lehoczky, C. Li, D. Knuteson, B. Raghothamachar, M. Dudley, J. Szoke and P. Barczy, “Crystal Growth of CdTe by Gradient Freeze in Universal Multizone Crystallizator (UMC)”, Materials Science Forum Vol. 508, Solidification and Gravity IV, 117 – 124 (2006).

Presentations:

1. C. Li, Ching-Hua Su, S. L. Lehoczky, R. N. Scripa, H. Ban and B. Lin, “Thermophysical Properties of Selected II-VI Semiconducting Melts”, presented at the Fourth International Conference on Solidification and Gravity (SG04), Miskolc-Lillafüred, Hungary, September 6-9, 2004.

2. Ching-Hua Su, S. L. Lehoczky, C. Li, D. Knuteson, B. Raghothamachar, M. Dudley, J. Szoke and P. Barczy, “Crystal Growth of CdTe by Gradient Freeze in Universal Multizone Crystallizator (UMC)”, ibid.

3. C. Li, S.L. Lehoczky, Ching-Hua Su and R. N. Scripa, "Electrical Conductivity of HgTe at High Temperatures", presented at 2004 Materials Research Society (MRS) Fall Meeting, Boston MA, November 29-December 3, 2004.

4. Chao Li, Ching-Hua Su, Sandor L. Lehoczky and Rosalia N. Scripa, "Impurity Studies of Cd0.8Zn0.2Te Crystals Using Photoluminescence and Glow Discharge Mass Spectroscopy", presented at the 16th American Conference on Crystal Growth and Epitaxy (ACCGE 16) held in Big Sky Resort, Montana, July 10-14, 2005.

II. Structural Transformation of Group I-III-VI Melts

The objective of the project is to conduct ground-based experimental and theoretical research on the structural fluctuations and thermophysical properties of molten II-VI compounds to enhance the basic understanding of the existing flight experiments in microgravity materials science programs and to study the fundamental heterophase fluctuations phenomena in these melts by:

1) Conducting neutron scattering analysis and measuring quantitatively the relevant thermophysical properties of the melts such as viscosity, electrical conductivity, thermal diffusivity and density as well as the relaxation characteristics of these properties to advance an understanding of the structural properties and the relaxation phenomena in these melts and 2) Performing thermodynamic analyses on the melts to interpret the experimental results.

During last year, the followings have been accomplished:

Publications in Refereed Journals:

1. Douglas. A. Barlow, James K. Baird and Ching-Hua Su, “A Theory of the von Weimarn Rules Governing the Average Size of Crystals Precipitated from a Supersaturated Solution”, J. Crystal Growth, 264 417-423 (2004).

2. Heng Ban, Bochuan Lin, Chao Li, Rosalia Scripa, Ching-Hua Su and Sandor L. Lehoczky, “Torque Transient of Magnetically Driven Flow for Viscosity Measurement”, Proceedings of 2004 Heat Transfer/Fluids Engineering Summer Conference, (HT/FED’04), HT-FED2004-56715 (2004).

3. C. Li, H. Ban, R. N. Scripa, Ching-Hua Su and S. L. Lehoczky, “Transient Torque Method: a Fast and Non-intrusive Technique to Simultaneously Determine Viscosity and Electrical Conductivity of Semiconducting and Metallic Melts” Review of Scientific Instruments, 75 2810-2816 (2004).

Presentations:

1. Bochuan Lin, Shen Zhu, Heng Ban, Chao Li, Rosalia Scripa, Ching-Hua Su and Sandor L. Lehoczky “Thermal Property Measurement of Semiconductor Melt Using Modified Laser Flash Method”, presented at 2003 Summer ASME Heat Transfer Conference, Las Vegas, Nevada July 21-23,2003.

2. Bochuan Lin, Shen Zhu, Heng Ban, Chao Li, Rosalia Scripa, Ching-Hua Su and Sandor L. Lehoczky “Modified Laser Flash Method for Thermal Property Measurements and the Influence of Heat Convection”, presented at 2003 ASME International Mechanical Engineering Congress and R & D Expo, Washington, D. C., November 15-21,2003.

3. Heng Ban, Bochuan Lin, Chao Li, Rosalia Scripa, Ching-Hua Su and Sandor L. Lehoczky, “Torque Transient of Magnetically Driven Flow for Viscosity Measurement”, presented at 2004 Heat Transfer/Fluids Engineering Summer Conference, Charlotte North Carolina, July 11-15, 2004.

4. C. Li, H. Ban, B. Lin, R. N. Scripa, Ching.-Hua Su, S.L. Lehoczky, “Temperature Dependence of Density, Viscosity and Electrical Conductivity for Hg-Based II-VI Semiconductor Melts”, presented at 14th International Conference on Crystal Growth (ICCG-14), Grenoble, France, August 9 –13, 2004.

5. H. Ban, C. Li, B. Lin, K. Emoto, R. N. Scripa, C.-H. Su, S.L. Lehoczky, “Thermal Diffusivity for III-VI Semiconductor Melts at Different Temperatures”, ibid.

6. C. Li, Ching-Hua Su, S. L. Lehoczky, R. N. Scripa, H. Ban and B. Lin, “Thermophysical Properties of Selected II-VI Semiconducting Melts”, presented at the Fourth International Conference on Solidification and Gravity (SG04), Miskolc-Lillafüred, Hungary, September 6-9, 2004.

7. C. Li, S.L. Lehoczky, Ching-Hua Su and R. N. Scripa, "Electrical Conductivity of HgTe at High Temperatures", presented at 2004 Materials Research Society (MRS) Fall Meeting, Boston MA, November 29-December 3, 2004.

9. Chao Li, Ching-Hua Su, Sandor L. Lehoczky, Rosalia N. Scripa and Heng Ban, "Thermophysical and Optical Properties of Semiconducting Ga2Te3 Melt" presented at the 16th American Conference on Crystal Growth and Epitaxy (ACCGE 16) held in Big Sky Resort, Montana, July 10-14, 2005.

III. Neutron Scattering Experiments:

1. A neutron diffraction experiment was performed for liquid Tellurium at Argonne National Laboratory, Intense Pulsed Neutron Source (IPNS), using the Glass, Liquid, and Amorphous Materials Diffractometer (GLAD), from December 6-14, 2004. The diffraction spectra were taken at sample temperature ranged from melting point 450 to 750oC with the addition of a supercooled temperature of 430oC. The data are being analyzed to examine the possible change of liquid structure from two- to three-fold domain at around 600oC.

2. A neutron scattering experiment on HgZnTe and Te melts was performed at IPNS at Argonne National Laboratory from August 20 to 26, 2003. The experiment was awarded to study the phase structure and transformation in molten HgZnTe and Te to support the ground-based project. The preliminary results showed a time-dependent drift of the scattering spectra even after 10 hours at constant temperature.

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| **Bibliography Type:** | Description: (Last Updated: 02/18/2025) |
| **Abstracts for Journals and Proceedings** | Li C, Su C-H, Lehoczky SL, Scripa R. "Impurity Studies of Cd0.8Zn0.2Te Crystals Using Photoluminescence and Glow Discharge Mass Spectroscopy." 16th American Conference on Crystal Growth and Epitaxy (ACCGE 16) held in Big Sky Resort, Montana, July 10-14, 2005.<br>16th American Conference on Crystal Growth and Epitaxy (ACCGE 16), July 2005. , Jul-2005 |

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| **Abstracts for Journals and Proceedings** | Li C, Su C-H, Lehoczky SL, Scripa R, Ban H. "Thermophysical and Optical Properties of Semiconducting Ga2Te3 Melt." 16th American Conference on Crystal Growth and Epitaxy (ACCGE 16) held in Big Sky Resort, Montana, July 10-14, 2005.<br>16th American Conference on Crystal Growth and Epitaxy (ACCGE 16), July 2005. , Jul-2005 |
| **Abstracts for Journals and Proceedings** | Su C-H, Lehoczky SL, Fishman GJ, Groza M, Burger A. "Growth of CdZnTe Crystals for Radiation Detector Applications using the Bridgman Technique with a Controlled Overpressure of Cd." SPIE (The International Society for Optical Engineering) Symposium on Optics & Photonics, San Diego, California, August 14, 2006.<br>SPIE (The International Society for Optical Engineering) Symposium on Optics & Photonics, August 2006. , Aug-2006 |
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| **Papers from Meeting Proceedings** | Su CH, Lehoczky SL, Li C, Knuteson D, Raghothamachar B, Dudley M, Szoke J, Barczy P. "Crystal Growth of CdTe by Gradient Freeze in Universal Multizone Crystallizator (UMC)." 4th International Conference on Solidification and Gravity, held in Miskolc-Lillafu¨red, Hungary, September 6-9, 2004.<br>In: Solidification and gravity IV : proceedings of the 4th International Conference on Solidification and Gravity, held in Miskolc-Lillafu¨red, Hungary, September 6-9, 2004 . p. 117 – 124 . Uetikon-Zuerich, Switzerland ; Enfield, NH : Trans Tech Publications, c2006. (Materials Science Forum ; Vol. 508) , Feb-2006 |
| **Papers from Meeting Proceedings** | Barczy P, Szoke J, Su C-H, Lehoczky SL, Schweizer M, Volz MP, Szofran FR. "Single Crystal Growth – New Technologies on Old Base" 4th International Conference on Solidification and Gravity, held in Miskolc-Lillafu¨red, Hungary, September 6-9, 2004.<br>In: Solidification and gravity IV : proceedings of the 4th International Conference on Solidification and Gravity, held in Miskolc-Lillafu¨red, Hungary, September 6-9, 2004 . p. 479-484. Uetikon-Zuerich, Switzerland ; Enfield, NH : Trans Tech Publications, c2006. (Materials Science Forum ; Vol. 508) , Feb-2006 |
| **Papers from Meeting Proceedings** | Lin B, Ban H, Li C, Scripa R, Su C-H, Lehoczky SL. "Laser Flash Method for Non-Opaque Materials." 2005 Summer Heat Transfer Conference (HT-2005), San Francisco, CA, July 17-22, 2005.<br>Proceedings of the 2005 Summer Heat Transfer Conference (HT-2005), ASME HT2005-72020 (2005). ISBN #: 0791847349 , Jul-2005 |
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